

Michael Bobak

Knowledge-Engineer / Research-Programmer

mike.bobak@gmail.com

linkedin.com/in/michaelbobak

[Champaign, IL](#)

[Tweets:@MBstream](#)

SUMMARY

[Research-Programmer](#) starting with (bio)[physical-science](#) simulation, adding AI study and years of [Knowledge-Engineering](#) work
[Knowledge-Based](#) aids, for process improvement to [teaching](#). AI: [Knowledge-Representation&Reasoning](#), [Rules](#), [Kn-Acq](#), [NLP](#), [ML](#), ...

WORK EXPERIENCE

Present
 Apr 2026

PivotPathSolutions *remote*

Subject Matter Expert

SME for digital-twin efforts, which incl. the use of semantics. First for a DARPA medical digital-twin but then for a general path forward

Dec 2025
 Dec 2023

Xendat/Vertex *remote*

[Ontology Data Developer/Analyst \(Integrations\)](#)

Contract ontology engineer through a shop serving a large pharma using: Python, SPARQL, Triple-Stores, [\(web\)](#)Protege, SemanticWeb-Libs

Wrote new libs used for both a Semantic-ETL and for an automated QA/QC, aiding Ontologist & BizDev, incl several apps

2025
 Jun 2018

AlohaHealth *remote*

[Sr AI Research-Engineer](#)

Part of a startup built on the topic of my UCSF research. Semantic search for clinical trials. (graph store/s) pivot incl: Intelligent document processing, of pharma CofAs etc, automation that also allows for better analytics

Aug 2023
 Nov 2019

National Center for Supercomputing Applications *Urbana, IL*

[Sr Research Software Engineer](#)

- All the PoCs for earthcube.org/geocodes incl. organizing others to bring in a great NSF review

- Focus on semantics/metadata search, with some NLP and sim. Using: Python, SPARQL, triple-stores, [FAIR data principles](#)

Aug 2019

Aug 2018

Agribile/Nutrien Urbana, IL[Sr Engineer, Natural Systems](#)

Ag-informatics/sim/.. Planned and guided reworking the main simulation, documentation and ML/verification (Python)

Jul 2018

Jul 2011

Freelance San-Francisco, CA

consultant

- Working as an ontologist for osthus.com on aligning bio/pharma ontologies to BFO to annotate masses of data in HDF5 files, for the [allotrope.org](#)
- Worked with IDEO on their systems integration issues that could be aided by Knowledge-Graph for information refinement and cleanup
- Worked with the Siemens Web of Things research group on use of SemWeb+IoT for adaptable manufacturing
- Advised start-ups in AI: fashion blog to trends, sport dbpedia info, work chatbot
- Developed ideas to take my UCSF research and fuse it with the Patient Data Mining Cluster that was developed by the UCSF Head of Research Computing and a PhD student, which has now been submitted for a patent
- Worked with UCSF in Psychology Department understand how to apply NLP and graph relation insights into an app they developed called Prime, to help with mental health management
- Continued to build skills around ML, Semantic-Web/Linked-Data, and Knowledge-Engineering, with these courses, from:
 - Coursera: Data Analysis, Data Science (with distinction), Machine Learning (with distinction), Discrete Optimization (audit)
 - openHPI: Semantic Web, Knowledge Engineering, Kn Eng w/Semantic Web technology, LinkedDataEngineering
 - Stanford: Design Thinking

Jul 2011

Oct 2010

ApolloGrp.edu San-Francisco, CAArchitect , [Adaptive Learning Platform](#)

Conceptually annotate study material & tests for automated remediation, instrument classroom to learn from use (Hadoop, Lisp, KM)

Oct 2010

Sep 2007

UCSF.edu San-Francisco, CA

Programmer/Analyst III

Medical-Informatics [research](#) (relating to clinical-trials) in Lisp/KM, and Natural-Language-Processing in Java/etc; [paper](#) with Stanford [group](#); [ontology](#) dev/use (Protege, Lisp, KM, ..)

Sep 2007

Feb 2001

Freelance Chicago/Boston

Knowledge-Engineer/ Research-Programmer

mindbox.com 3/02-10/02. (used Art*Enterprise) See: Ocwen_Mindbox Worked up to half-time for cas.dis.anl.gov 5/03-5/04 (Java Simulation) Worked full-time 8/03--05(verizonlabs.gte.com, Model-Based-Diagnosis on a national scale. (Art *Enterprise) See: aaai.org/Papers/IAAI/1996/IAAI96-287.pdf Bioinformatics/control contract 11/04-12/05 (CLIPS&Protege.stanford.edu/Java/DB) Control of perfusion pumps on light microscope sample, monitoring incl. Machine-vision, Bio-ontology/reasoning/Kn-mngt for the experiment setup. & Grant proposal work. Worked for CME.com 2/06-06/06 (re)organizing trade-data validation code. (CLIPS/Jess) Signal-Processing/Machine-Learning (startup) 06/06-(Lisp/etc) Hospital Informatics/Machine-Learning ghx.com 02/07-05/07-(Lisp), MachineLearning speedup for financial-scientific (Lisp)

Feb 2001

Jun 1998

kbs.ai.UIUC.edu Urbana, IL(Senior) Research Programmer (Knowledge Based Systems Lab)

University of Illinois Urbana-Champaign, IL Organize many levels of a very large knowledge based simulation projects. Brought over 18 programmers together to deliver a coherent product. Ran weekly (sub)group meetings, down to help solving any problem. Hiring, demo, design, install trips, prototyping to lead project direction. Taught group of 6 how to use a Rule-Based-shell for a reasoner-rewrite in Art*Enterprise. Projects included: Simulation-based, Intelligent Tutoring System (ITS) & Real-Time control system. Being used in classroom, real life testing, presented at IAAI99 'Automated Instructor Assistant for Ship Damage Control' The system teaches Navy officers how to save a simulated ship in crisis. A variant was developed to catch real-time crisis conditions and suggest solutions www.dwilkins.org/members.htm

Jun 1998

Oct 1996

Brightware out of Chicago, IL

Knowledge-Engineer

Helped develop and install their very [first product \(Intelligent email reply\)](http://first product (Intelligent email reply)). Worked between development and consulting. Helped on several Knowledge-Based business applications. Helped with several deployed Knowledge-Based business applications (ie. financial: mortgage, web based job finder). (Art*Enterprise)See: http://www.brightware.com/eservice_solutions/ More recently I worked 1/2year for the new version of the company: Mindbox.

Aug 1996

Feb 1996

Institute of Learning Sciences Evanston, IL

Lead Programmer/Analyst

Wrote Lisp code (mainly GUI) for Qualitative Research Group. Learned more about Qualitative/Quantitative Simulation, Model-Based Reasoning, Intelligent-Tutoring-Systems, & general Lisp programming. See: <http://www.qrg.northwestern.edu/projects/NSF/Cyclepad/aboutcp.htm>

Feb 1996

Feb 1993

Argonne National Lab Argonne, ILSoftware Engineer (EAD then DIS groups)

Wrote fielded Expert System by myself at the end of grad-school. (in Lisp rule-shell then CLIPS) Prototyped communication & control of distributed simulation. (in CLIPS PVM etc) Agent wrapping of simulations with

CLIPS+PVM, to describe then mix and match them. Also used C++/Smalltalk/FORTRAN with PVM; Other work as needed. Algo/Viz/Etc. Written up in a book about innovative distributed object application. See: http://www.dis.anl.gov/DEEM_HLAsim http://www.dis.anl.gov/DEEM/DIAS_diaswp.pdf _More recently I worked part-time for the new subgroup of dis: cas.dis.anl.gov.

Jan 1993

Jan 1990

UIUC.edu *Urbana, IL*

Graduate Research Assistant /Research Programmer

Wrote molecular graphics package used in classes & for publications. (in C) Used machine-learning techniques for protein structure prediction.

Wrote thesis on Knowledge-Based Simulation Environment. (Lisp/OPS5/C) Overseen by heads of the NCSA CompBio group and head of Biophysics at the time. see: web.bilkent.edu.tr/ncsa/Apps/CBdir.html

Dec 1989

Apr 1989

National Center for Supercomputing Applications(NCSA)Uof IL,GIST *Urbana/Savoy, IL*

Programmer/Consultant

Suggested scientific software path for Software Tools Group of NCSA; Wrote molecular viz code for a professor. Wrote testing code for Global Info Systems Tech. (in C)

Apr 1989

Oct 1988

Shearson Lehman Hutton *London, England*

Programmer (Research Computing)

Maintained financial databases & daily report information. Organized worldwide mailing system. Wrote statistics code for stock predictions. (MUMPS and Maths-package)

Aug 1988

Mar 1982

US Army Corp. of Eng. Research Lab *Champaign, IL*

Research Programmer (Modeling then Acoustics teams)

Provided research support from start to finish. (FORTRAN) Wrote and ran computer simulation code, compared output with field data. Did field measurements to back up predictions. (Team/Self; Local/US/World-wide) My work went into several [published papers](#). GRASS: grass.osgeo.org

EDUCATION**University of Illinois, Urbana-Champaign**

MS Biophysics & Computational Biology with AI, 1990-93

BS Physics, BS Biophysics, 1983-88, dept. distinction

PROFESSIONAL ORGANIZATIONS:

[AAAI \(Association for the Advancement of Artificial Intelligence\)](#) life-member.
[IEEE \(Institute of Electrical and Electronics Engineers\)](#) & [Computer Society](#) 10yr

Other groups:[meetup.com](#), [linkedin-groups](#)**ID**

orcid.org/0000-0003-2357-5918
wikidata.org/wiki/Q104512704

Papers

scholar.google.com/&q=michael+bobak

SKILLS & EXPERTISE

AI [Artificial Intelligence](#) [Adaptive Systems](#) [Business Rules](#) [Case-Based Reasoning](#)

[Composite Applications](#) [Computational Intelligence](#) [Conceptual Modeling](#) [Controlled Vocabularies](#) [Causal Inference](#)

[Data Analysis](#) [Data Quality](#) [Data Mining](#) [Decision Modeling](#) [Expert Systems](#) [Information Access](#) [Information Extraction](#)

[Information Retrieval](#) [Intelligent Agents](#) [Intelligent Systems](#) [Intelligent Tutoring Systems](#) [Knowledge Acquisition](#)

[Knowledge-based Systems](#) [Knowledge Engineering](#) [Knowledge Representation](#) [Logic Programming](#) [Mathematical Logic](#)

[Mathematical Programming](#) [Machine Learning](#) [Metadata standards](#) [Model-based reasoning](#) [Natural Language Processing](#)

[Natural Language Understanding](#) [Ontology Development](#) [Ontology Engineering](#) [Recommender-Systems](#) [Rules](#) [Rules Engines](#)

[Semantics](#) [Semantic Search](#) [Semantic Technologies](#) [Semantic Web](#) [SNOMED](#) [Taxonomy Development](#) [Text Classification](#)

Science [Research](#) [Scientific Software](#) [Scientific Computing](#) [Scientific Visualization](#) [Simulation](#)

[Computational Mathematics](#) [Biophysics](#) [Computational Biology](#) [Physics](#)

Others [Cloud Computing](#) [MapReduce](#) [Hadoop](#) [Dynamic Languages](#) [Exploratory programming](#)

[Common Lisp](#) [other Languages](#)

PROGRAMMING LANGUAGES/...:

25+ years
overall

Object Orientated
14+yr

Rule-Based
KnRep&
Reasoning:10+yrs

Libs:

Databases:

Operating-Systems:

C 6+yrs FORTRAN
6+yrs

[Smalltalk](#)(1yr),C++ 1+yr

[OPS5](#), [Prolog](#),[GoldWorks](#) 1yr [Viz: OpenGL](#) 3+yrs

MS-Jet/SQL, MySQL NeXTSTEP, MS(NT..XP)
(8+yr)

Scheme ~1 yr MUMPS
1/2yr..

Python 8+yrs

[CLIPS](#) 4+yrs,[ART](#)

[HPC: PVM](#) 1+yr

PostgreSQL, [ORDB](#) UNIX/[GNU/Linux](#)
25+yrs

[CommonLisp](#) 7+yrs

[CLOS](#)(CL -Object-
System)

[Knowledge-Machine](#) 3+yrs

[WS](#):Tomcat/Axis
SOAP/REST

[Graph&triple](#) 8+yrs [OS-X](#).Darwin 20+yrs

[other Lisps](#) 10+yrs

Java/Scala 2+yrs

[JESS](#)1yr,[Protege](#) 8+yrs

[persistance&NoSQL](#)

[some opensource examples at: github.com/MBcode .github.io](#)

as [html/ pdf](#) and [latest-talk](#) ..